

Question Paper

Exam Date & Time: 30-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. DEGREE EXAMINATIONS - DECEMBER 2025

High Performance Computing Systems [CSS_5103]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Evaluate in detail the commonly used OpenMP built-in functions for: (5)
- A) i. Thread identification
ii. Timing
iii. System information
Include syntax and example usage.
Develop a OpenMP program to perform multiplication of two $N \times N$ matrices ($N \leq 500$) using `#pragma omp parallel` for parallelization. Display the resulting matrix C.
- Course Outcome: "CO2" / Cognitive Level: "Analysis"
- B) Why do we need Parallelism? Demonstrate the following parallel computer structures with diagram. (3)
- i. Pipeline computer
ii. Array computer
iii. Multiprocessor systems
- Course Outcome: "CO1" / Cognitive Level: "Application"
- C) Explain temporal parallelism and spatial parallelism. (2)
- Course Outcome: "CO1" / Cognitive Level: "Remember"
- 2) Develop a MPI program to read N elements of the array in the root process (process 0) where N is (4)
- A) equal to the total number of processes. The root process sends one value to each of the slaves. Let even ranked process find square of the received element and odd ranked process find cube of received element. Use Buffered send.
- Course Outcome: "CO3" / Cognitive Level: "Create"
- B) Demonstrate the Parallel Merge Sort algorithm to sort N elements using tree allocation of processes (3)
- Course Outcome: "CO3" / Cognitive Level: "Application"
- C) Analyze the working of `MPI_Allgather()` in an MPI program where each process contributes a (3)
- unique value, and all processes receive the complete set of values.
Discuss how it differs from `MPI_Gather()`.
- Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Solve the problem of reversing a given string *str* with length *len=5* and store the resultant string to (5)
- another character array *rev_str* by developing an OpenCL code snippet with the help of kernel

- A) function *sine*. Develop only the code snippets starting from kernel object creation till reading the result from the device to the local variable *strres*. Note: Assume that other required objects are already created.

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) Determine how to find out the execution time taken by a OpenCL kernel function starting from event object creation. Specify all the statements required for this. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- C) Explain workgroup and work item by taking the example of a vector of 16 workgroups and 32 work items in each workgroup. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4) Construct a parallel algorithm to sum n values using Shuffle-exchange SIMD model where n is the number of values to be added and p is the number of processors in the model. It is assumed that each processor initially holds one of the values among n . (4)

A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Demonstrate the Centralized shared-memory multiprocessor and Distributed memory multiprocessor (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Discuss the node degree and the network diameter metrics of a static interconnection network of SIMD computers. (3)

Course Outcome: "CO6" / Cognitive Level: "Understand"

- 5) Develop a CUDA program to read a matrix A of size $N \times N$. It replaces the principal diagonal elements with zero. Elements above the principal diagonal by their factorial and elements below the principal diagonal by their sum of digits. (4)

A)

Course Outcome: "CO5" / Cognitive Level: "Create"

- B) Determine the CUDA runtime system API functions for managing the data in device memory. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

- C) Develop CUDA kernel to accept an integer array of size `arr_size`, that is multiples of 32. Each thread finds the average of consecutive 32 elements in the input array and stores it in an output array in the appropriate position. Also, write the block size used. (3)

Course Outcome: "CO5" / Cognitive Level: "Create"

-----End-----

Event Name	Subject
058-MIT BLR-MAKEUP EXAM PG-2025-2026	CSS_5103:High Performance Computing Systems

#	Course Outcome	Description	Marks
1	1	CO 1	12
2	2	CO 2	5
3	3	CO 3	10
4	4	CO 4	10

5	5	CO 5	10
6	6	CO 6	3
			Total 50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-MIT BLR-MAKEUP EXAM PG-2025-2026

CSS_5103:High Performance Computing Systems

#	Level	No of Questions	Marks (%)
1	Remember	1	2(4%)
2	Understand	2	5(10%)
3	Application	7	24(48%)
4	Analysis	2	8(16%)
5	Create	0	0(0%)
6	Evaluate	3	11(22%)
Total		15	50 (100%)